One computational system or two? Evidence from semantically incongruent propositions

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BACKGROUND. Bimodal bilinguals, i.e., competent users of one spoken and one signed language, provide a unique opportunity to investigate the neural architecture and cognitive processing mechanisms of the bilingual mind **[5,4]**, given their extraordinary capacity to produce two utterances simultaneously in two different modalities, namely speech and sign. In doing so, bimodal bilinguals have been shown to occasionally produce syntactically incongruent, that is, mismatching word orders (1). This finding has sparked a debate in the psycholinguistics literature over whether bimodal bilinguals have a single computation / derivation **[7,6]** that generates two word orders, or separate, simultaneous computations or derivations **[2,1]**, one per word order. A better way to address the above debate likely comes from examining semantically incongruent sentences, especially when participants are prompted to do so.

OBJECTIVE. This study aimed to examine whether bimodal bilinguals are in fact capable of producing semantically incongruent sentences in speech and sign by investigating a bimodal bilingualism context involving the newly emergent *Sadat Tawaher Sign Language (STSL)* and the surrounding spoken language, *Khuzestani Arabic (KhA)*. STSL emerged naturally around sixty years ago within a single family in the southwestern Iranian village of *Sadat Tawaher*, following a case of sudden, complete hearing loss. To date, STSL has been learned by three generations, with almost all signers regularly speaking while signing.

METHODOLOGY. The data for the current study were collected from 13 hearing KhA-STSL fluent bimodal bilinguals (9 males, 4 females), with a mean age of 31 years (age range: 7-56 years). Most signers had acquired STSL in early childhood, and four were CODAs. The data were gathered in two steps. In the first step, participants produced 1,100 sentences through a sentence production task and a free narrative elicitation task, with both tasks conducted off-line. All sentences were produced simultaneously in speech and sign and included various types, such as affirmative declaratives, negatives, and wh-questions. In the second step, participants were asked to simultaneously produce two semantically incongruent sentences (e.g., '*I'm not hungry*' in speech and '*I'm full*' in sign) during live sessions.

FINDINGS & DISCUSSION. The results from the first step of data collection showed that almost all sentences involved code-blending, i.e., the simultaneous production of a word and a sign. Moreover, nearly 33% of sentences exhibited incongruent (i.e., mismatching) word orders in speech and sign. Code-switching was observed only occasionally and occurred mostly from sign to speech. However, no semantically incongruent sentences were produced in this step. These findings are consistent with previous research on bimodal bilingualism in other contexts, e.g., ASL-English **[3,4]**, Italian Sign Language-Italian **[2]**, among others. The second step of data collection was far more revealing: participants failed to produce any semantically incongruent propositions. They reported finding the task extremely challenging, and often either code-blended both propositions or resorted to code-switching. None of the participants completed the task successfully.

CONCLUSION. The non-occurrence of semantically non-equivalent propositions in sign and speech, combined with STSL–KhA signers' inability to produce such structures, strongly suggests that the bimodal bilingual mind is critically limited in encoding two semantically distinct propositions due to their high cognitive cost, and that such use of the two modalities is generally avoided. This limitation is likely attributable to memory or processing constraints. If so, bimodal bilinguals, despite having two output channels–vocal tract and hands–appear to have a single computational system that sometimes generates utterances in two different surface word orders **[6],** but never two semantically incongruent propositions. In this regard, semantically non-equivalent propositions impose the greatest cognitive cost on the production system.

Examples:

(1) **KhA**: ət̪ʃ-t̪ʃæləb tˤəfær ən-næhær **(SVO)**
the-dog jumped.3sg the-creek

STSL: DOG CREEK JUMP **(SOV)**
'The dog jumped over the creek.'

Semantically incongruent sentences:

(1) Speech: I am not hungry.
Sign: I am full.

(2) Speech: My friend didn't buy a car.
Sign: My friend saved his money.

(3) Speech: He stayed home.
Sign: He didn't go to work.

(4) Speech: He's single.
Sign: He's not married.

(5) Speech: It rained yesterday.
Sign: It was cold yesterday.

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